ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 2 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals paper 2 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, examples and equivalent code.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 C ✓ 1.1.4 A ✓ 1.1.5 B ✓ 1.1.6 D ✓ 1.1.7 A ✓ 1.1.8 C ✓ 1.1.9 C ✓ 1.1.10 A ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 B ✓ 1.3.2 H ✓ 1.3.3 J ✓ 1.3.4 G ✓ 1.3.5 C ✓ 1.3.6 D ✓ 1.3.7 F ✓ 1.3.8 I ✓ 1.3.9 E ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Design (model in CAD) ✓ → file processing (export STL/3MF and slice) ✓ → production (printing) ✓ → finishing (remove supports, clean, sand) ✓ (4)

2.2 PLA: lower temperature (about 190–220 °C) ✓, less heat resistant / more brittle ✓, easy to print, little warping, no heated bed needed ✓; ABS: higher temperature (about 230–250 °C), stronger and more heat resistant ✓, warps easily, needs a heated bed and gives off fumes ✓. Recommendation with reason, e.g. PLA – easy and safe for a small cover; or ABS if the battery gets warm (accept either if motivated) (5)

2.3 Any FOUR ✓✓✓✓: extruder / feeder – pushes the filament; hot end / nozzle – melts and deposits the plastic; print bed – surface the object is built on; stepper motors / axes – move the nozzle and bed in X, Y and Z; controller board – reads the G-code and controls the printer; cooling fan – cools each layer (4)

2.4 Advantages (any TWO) ✓✓: custom parts quickly and cheaply; complex shapes possible; easy to change the design; little waste. Disadvantages (any TWO) ✓✓: slow for large parts; plastic parts are weaker than metal; limited size; layer lines / needs finishing; printing faults (warping) (4)

2.5 If the first layer does not stick, the part moves or warps and every layer above fails ✓✓; improve adhesion by levelling the bed / cleaning the bed / using a heated bed / glue stick or tape / a brim or raft (any ONE) ✓ (3)

[20]

QUESTION 3

3.1 Voltage is like water pressure that pushes the water ✓; current is like the amount of water flowing through the pipe ✓; resistance is like a narrow pipe that restricts the flow ✓. (3)

3.2.1 $V_R = 5 - 2 = 3\ V$ ✓ (1)

3.2.2 $R = V \div I$ ✓ $= 3 \div 0{,}02$ ✓ $= 150\ \Omega$ ✓ (3)

3.2.3 $P = V \times I$ ✓ $= 3 \times 0{,}02 = 0{,}06\ W\ (60\ mW)$ ✓ (2)

3.3 (a) Voltage: multimeter in parallel across the LED ✓ (b) current: break the circuit and connect the multimeter in series ✓ (2)

3.4 With the power off, set the meter to continuity and touch the probes along the circuit ✓; where the meter does not beep there is a break (open circuit), e.g. a loose wire or broken joint ✓. (2)

[13]

QUESTION 4

4.1 Arduino Uno: microcontroller board ✓, no operating system – runs one sketch ✓, simple real-time control of sensors and motors ✓; Raspberry Pi: single-board computer / microprocessor ✓, runs an operating system (Raspberry Pi OS / Linux) ✓, cameras, networking, more complex programs and AI ✓ (6)

4.2 Each short row of five holes on either side of the centre gap is connected ✓; the two halves are not connected across the centre gap ✓; the long power rails along the sides are connected along their length (+ and −) ✓. (3)

4.3 Series: the circuit is broken, so the other LED also goes off ✓✓. Parallel: the other LED still has its own complete path and stays on ✓✓. (4)

4.4 (a) PIR motion sensor ✓ (b) pressure pad / force-sensing resistor (FSR) ✓ (c) photoresistor (LDR) / light sensor ✓ (3)

4.5 The Raspberry Pi GPIO pins are digital only (on/off) ✓; the ADC converts the varying analogue voltage into a number the Pi can read ✓. (2)

4.6 Passive: cannot amplify or control power, needs no extra power to work ✓, e.g. resistor, capacitor ✓; active: can control / amplify current or needs a power source ✓, e.g. transistor, IC, LED ✓ (4)

[22]

QUESTION 5

5.1.1 LED on GPIO 17 ✓, button on GPIO 2 ✓ (2)

5.1.2 The LED switches on ✓ for 3 seconds and then switches off ✓. (2)

5.1.3 It makes the program repeat forever (an endless loop that keeps checking the button) ✓. (1)

5.1.4 Correct loop ✓, on ✓, off ✓, correct delays ✓, e.g. (4)

```
    if button.is_pressed:
        for i in range(5):
            led.on()
            sleep(0.5)
            led.off()
            sleep(0.5)
```

5.2 Components (any THREE) ✓✓✓: Raspberry Pi (or Arduino), push button, buzzer, LED, resistor, breadboard and jumper wires. Flowchart: start → wait / check if button pressed ✓ → buzzer on, LED on, wait 2 s, buzzer off ✓ → wait 8 s more, LED off, back to checking ✓ (6)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE